

Uncovering genetic clues to breast cancer chemo response



Nucleic acid quantification using Implen NanoPhotometer

Accurate quantification of nucleic acids is crucial in cancer research for number of reasons. From detection and analysis of genetic alterations to monitoring treatment response, quantification of DNA, RNA and nucleic acids has its own role in advancing our comprehension of cancer research. This application sheds lights on the use of Implen NanoPhotometer in DNA and RNA quantification in the context of breast cancer research, focusing its role in understanding genomic changes, identifying biomarkers and developing targeted therapies.

Keywords: DNA, RNA, quantification, neoadjuvant chemotherapy, breast cancer, Implen NanoPhotometer

Introduction

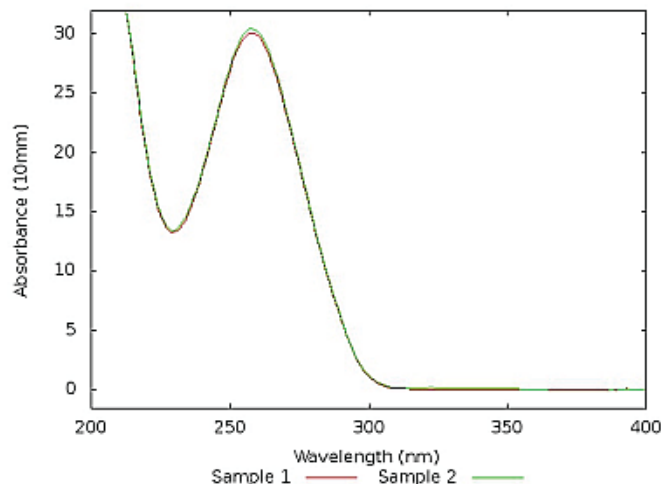
Neoadjuvant chemotherapy is a systematic pre-surgical treatment where chemotherapy is given before surgery for patients with localised high risk or inoperable tumours. Often used for breast cancer, it is aimed at downsizing the tumour to avoid axillary dissections in patients. Successful therapy paves way for preservation of organs, avoiding mastectomy, and reducing surgical dissections in sensitive patients. Although there are studies which indicate significant advantages of neoadjuvant chemotherapy, predicting its response is difficult due to tumour heterogeneity and limited precision of existing biomarkers. Therefore, the key goal remains achieving subsequent pathological complete response. To accomplish this, prediction of novel biomarkers by identifying patients who might benefit from neoadjuvant therapy.

This application highlights the use of Implen NanoPhotometer® for the quantification of DNA and RNA.

Application Showcase

Parameter									
Method	Nucleic Acids			Nucleic Acid Factor			40.00		
Type	RNA			Background Correction			320 nm		
Mode	NanoVolume			Air Bubble Recognition			Off		
Volume (ul)	1-2			Manual Dilution Factor			1.000		

#	Name	Conc.	Units	A230	A260	A280	A320	A260/A280	A260/A230	Dilution
1	Blank 1	0.0000	ng/ul	0.000	0.000	0.000	0.000	0.000	0.000	
2	Sample 1	1195.3	ng/ul	13.29	29.89	14.02	0.008	2.133	2.250	140
3	Sample 2	1203.1	ng/ul	13.42	30.18	14.23	0.103	2.129	2.259	140



Parameter									
Method	Nucleic Acids			Nucleic Acid Factor			50.00		
Type	dsDNA			Background Correction			320 nm		
Mode	NanoVolume			Air Bubble Recognition			Off		
Volume (ul)	1-2			Manual Dilution Factor			1.000		

#	Name	Conc.	Units	A230	A260	A280	A320	A260/A280	A260/A230	Dilution
1	Blank 1	0.0000	ng/ul	0.000	0.000	0.000	0.000	0.000	0.000	
2	Sample 1	795.20	ng/ul	6.993	15.90	8.405	-0.004	1.891	2.273	15

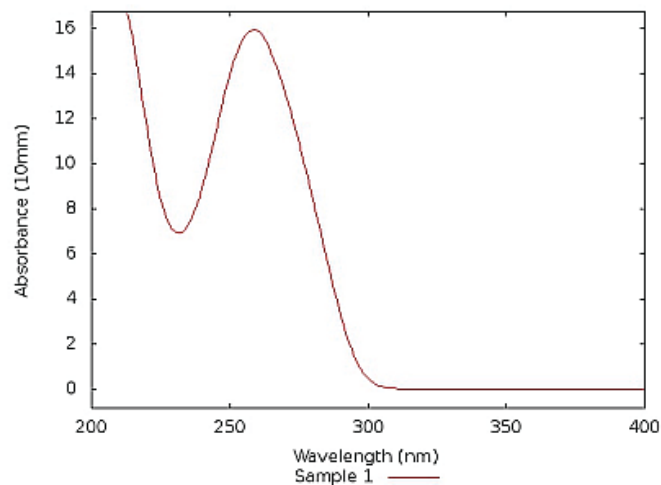


Figure 1: Example outputs of DNA and RNA quantification using Implen Nanophotometers®

Methods

Four datasets of patient populations were used in this study - NACBC Sequencing Set, Internal NACBC Validation Set, External GEO Validation Set (GSE25066) and External TCGA Validation Set, each of varying characteristics.

DNA was extracted from fresh frozen tissue samples and blood samples using the QIAamp DNA Mini Kit and QIAamp DNA Blood Mini Kit (Qiagen). RNA was extracted using TRIzol reagents (Tiangen).

Nucleic acid quantification

Purity Assessment: The purity of isolated DNA and RNA samples was evaluated using the NanoPhotometer®. The absorbance ratio of 260 nm to 280 nm (A260/A280) was determined. A ratio within the range of 1.8 to 2.0 indicated pure samples suitable for subsequent experiments.

Concentration Measurement: The concentration of nucleic acids was quantified based on the absorbance at 260 nm. A mass $\geq 3 \mu\text{g}$ was considered sufficient for sequencing sample library construction.

Conclusion

Key findings of this study indicate neoadjuvant chemotherapy significantly alters mutation rates, DNA repair pathways, and immune microenvironment. Specifically, the CDKAL1P409L mutation was identified as reducing chemotherapy sensitivity, while amplifications in ADGRA2 or ADRB3 were linked with worse prognosis and treatment outcomes.

The application note highlights the use of Implen Nanophotometer for DNA and RNA quantification. Specifically, after isolating total DNA from fresh frozen tissue samples and blood samples, and extracting RNA from fresh frozen tumour tissue, the purity of the total DNA and RNA was estimated using the NanoPhotometer.

Sample outputs of DNA and RNA quantification using the Implen NanoPhotometer is shown in Figure 1.

Reference

1. Yin, G.; Liu, L.; Yu, T.; Yu, L.; Feng, M.; Zhou, C.; Wang, X.; Teng, G.; Ma, Z.; Zhou, W.; Ye, C.; Zhang, J.; Ji, C.; Zhao, L.; Zhou, P.; Guo, Y.; Meng, X.; Fu, Q.; Zhang, Q.; Li, L. Genomic and Transcriptomic Analysis of Breast Cancer Identifies Novel Signatures Associated with Response to Neoadjuvant Chemotherapy. *Genome medicine* 2024, 16 (1). DOI: 10.1186/s13073-024-01286-8.



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